QQ-G-540a
JULY 27, 1956
SUPERSEDING
Int. Fed. Spec. QQ-G-00540 (GSA-FSS)
January 27, 1956, and
Fed. Spec. QQ-G-540
March 7, 1940

FEDERAL SPECIFICATION

# GOLD; CASTING, DENTAL

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies*

## 1. SCOPE AND CLASSIFICATION

1.1 Scope—This specification covers dental casting alloys based on gold and used in the fabrication of inlays, crowns, bridges and partial dentures (see 6.2 and 6.3).

1.2 Classification.

1.2.1 *Types.*—Dental casting gold shall be of the following types, as specified:

Type I.—Soft.
Type II.—Medium.
Type III —Hard.
Type IV.—Extra hard.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Specifications:*

U-I-546 — Investment; Casting, Inlay, Dental.
NN-B-621 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-601 — Boxes; Wood, Cleated-Plywood
PPP-B-676 — Boxes; Set-Up, Paperboard.

*Military Specifications:*

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
MIL-B-10377 — Boxes, Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

*Military Standard:*

MIL-STD-129 — Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other products specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo, Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.



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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications, Standards, Drawings, and Publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

**2.2 Other publications.** — The following publication of the issue in effect on date of invitation for bids, forms a part of this specification:

*American Dental Association Specification* No. 5—Dental Inlay Casting Gold Alloys.

(Copies of American Dental Association specifications may be obtained free upon application to the American Dental Association Research Fellowship, National Bureau of Standards, Washington 25, D C.).

## 3. REQUIREMENTS

**3.1 Material.** — The material shall be as specified hereinafter for each type.

**3.2 Color.**—The color shall be as specified in the invitation for bids.

**3.3 Castings.**—The molten gold shall fill the mold completely when cast into dental casting investment.

**3.4 Composition and physical properties.**—The physical properties shall be as given in table I for the respective types.

**3.5 Identification marking.**—Unless otherwise specified, each ingot shall be stamped with the name or registered trade-mark of the manufacturer or contractor and the last three numbers of the Federal Stock Number applicable to the particular item.

**3.6 Workmanship.**—Each ingot shall be of uniform composition and color throughout.

## 4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Examination for compliance with this specification.

**4.2 Composition.**—The determined values for metallic constituents shall be recorded at the nearest 0.5 percent. When a determined value falls midway between a half and a whole number, the whole number shall be recorded.

**4.3 Physical properties.**—Physical properties, other than fusion temperatures, shall be determined on cast samples placed in a furnace at 1,290°F. for 10 minutes and quenched in water at room temperature. Where hardened specimens are demanded these shall be prepared by placing quenched specimens in an oven heated to 840°F. After they have been

TABLE I.—*Composition and physical properties*

| Type | Gold and platinum[1] group metals, percent | Silver, percent | | Brinell number | | Elongation 2-inch gage length percent | | Strength (pounds per square inch) | | | Strain, inch per inch | Fusion temperature, degrees F. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Quenched | Hardened | Quenched | Hardened | Ultimate tensile | Yield | | | |
| | | | | | | | | | Quenched | Hardened | | |
| | *Min* | *Min.* | *Max.* | *Min.* | *Max.* | *Min.* | *Min* | *Min.* | *Min* | *Min* | *Max* | *Min.* |
| I | 90 | 3 | 10 | 40 | 75 | ---------- | 18 | ---------- | ---------- | ---------- | ---------- | 1,725 |
| II | 85 | 0 | 15 | 70 | 100 | ---------- | 12 | ---------- | ---------- | 22,000 | ---------- | 0.0017 | 1,650 |
| III | 80 | 0 | 15 | 90 | 140 | ---------- | 12 | ---------- | ---------- | 27,000 | ---------- | .0020 | 1,650 |
| IV | 75 | 0 | 15 | 130 | --- | 200 | 10 | 2 | 90,000 | ---------- | 65,000 | 0050 | 1,625 |

[1] Palladium is not permitted in excess of 4 percent in any of these alloys



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in the oven 2 minutes the temperature shall be uniformly reduced to 480°F. in 30 minutes. Cast samples shall be prepared by casting into a dental casting investment which complies with Federal Specification U–I–546, by any dental casting method. The use of a wax pattern is not required.

4.3.1 An alloy complies with this specification when all of the recorded values for the properties of three or more of a series of six castings meet the requirements. The recorded value for each property shall be obtained by averaging the determined values for all samples meeting the requirements of the specification.

4.4 For determining compliance with this specification, the successful bidder shall be required to supply six cast specimens with sprue and button attached for types I, II, and III, and 12 specimens with sprue and button attached for type IV at the time of delivery. These specimens shall be not less than 4 inches long and not less than 0.07 inch nor more than 0.09 inch in diameter. Agreement of the cast specimens in composition, purity, etc. with delivered material shall be determined by chemical analysis or such other methods as may be necessary.

4.5 **Brinell number.**—The Brinell number shall be determined by applying a load of 27.8 pounds (12.6 kg.) on the sample through a $\frac{1}{16}$ inch hardened steel ball for 30 seconds. The diameter of the depression shall be determined by measuring two diameters perpendicular to each other and the average diameter used in the calculation of the hardness number. The determined value shall be recorded as the nearest whole number. When the determined number falls midway between two numbers, the even number shall be recorded, (see par. 6 4).

4.6 **Yield strength.**

4.6.1 The yield strength shall be determined on cast samples whose diameters are between 0.07 and 0.09 inch. The value for yield strength shall be determined on a gage length of not less than 50 mm. (approximately 2 inches). An averaging extensometer shall be used.

4.6.2 An initial load calculated to produce a stress of 5,000 pounds per square inch shall be applied to the sample and the extensometer read. The load shall then be increased to the minimum value for the yield strength (type II, 22,000 pounds per square inch; type III, 27,000 pounds per square inch; and type IV, 65,000 pounds per square inch) and a second reading taken. The maximum strain permitted between the limits specified is type II, 0.0017 inch per inch; type III 0.0020 inch per inch; and type IV 0.005 inch per inch. The determined values for strain shall be recorded to the nearest 0.0001 inch per inch. When the determined strain falls midway between two numbers, the nearest even number shall be recorded.

4.6.3 As an alternate method, loads above and below the two limits specified may be applied provided they do not vary the stress over 500 pounds per square inch from the specified load. The recorded strain may be determined graphically.

4.7 **Elongation.**—Elongation shall be determined on the samples used in the test for yield strength. The load shall be applied slowly (not more than 20 minutes between initial load and rupture) and continuously after the extensometer is removed until rupture occurs. The load shall be applied slowly and continuously after the strain gage is removed until rupture occurs. Castings which break outside the gage marks may be repulled. Elongation shall be determined on a 50 mm. (approximately 2 inches) gage length. The determined value shall be recorded at the nearest 0.5 percent. When the determined value falls midway between a half and a whole number, the whole number shall be recorded.

4.8 **Fusion temperature wire method.**—The fusion temperature is the temperature at



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which a wire 0.70 to 0.80 mm. (0.028 to 0.032 inch) in diameter and approximately 2 mm. (0.08 inch) long, supported at both ends, will break when subjected to a cross bending load of 85 grams (approximately 3 ounces). The heating rate within 100°F. of the breaking point shall be between 5 and 28°C. (10° and 50°F.) per minute.

5. PREPARATION FOR DELIVERY (For definitions and levels see 6.1).

5.1 Preparation for delivery (Civilian).

5.1.1 *Packaging.*—Unless otherwise specified in the invitation for bids, castings shall be delivered in the form of ingots weighing 2 dwt. each. Each ingot shall be placed in an individual envelope and sealed. A standard of packing shall be required that conforms to Interstate Commerce Commission regulations and to good commercial practice.

5.1.2 *Packing.*—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

5.1.3 *Marking.*

5.1.3.1 *Package.*—The name of the contractor or manufacturer, name of the material, type, color, and minimum net weight of contents, shall be marked on each envelope, together with serial number or combination of letters and numbers which shall refer to the manufacturer's records for the particular lot or batch of alloy.

5.1.3.2 *Shipping container.*—Unless otherwise specified, shipping containers shall be marked on one end with the name of the material, the type, the color and the number of units contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

5.2 Preparation for delivery (Military).

5.2.1 *Packaging.*

5.2.1.1 *Unit of issue.*—One (1) ingot weighing 2 dwt. constitutes one unit.

5.2.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table II.

TABLE II.—*Packaging and packing quantities*

| Unit pack | Intermediate pack | Protective pack | Shipping container |
|---|---|---|---|
| 1 unit | 10 units | 100 units | 1,000 units |

5.2.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design acceptable to a common carrier, which will insure safe delivery to destination.

5.2.1.3 *Level A.*

5.2.1.3.1 *Unit package.*—Each unit shall be packaged in an envelope of suitable size and design Envelope shall be adequately secured to prevent accidental opening.

5.2.1.3.2 *Intermediate package.*—The number of unit packages specified in table II shall be wrapped with Kraft paper or banded together with a Kraft paper band Band or wrapping shall be adequately secured to prevent accidental opening.

5.2.1.3.3 *Protective package.*—The number of units specified in table II shall be packaged in a carton of appropriate size constructed in accordance with the requirements of Federal Specification PPP–B–566 or PPP–B–676. Carton shall be secured to prevent accidental opening.

5.2.1.4 *Level B.*—Packaging shall be as specified in paragraph 5.2.1.3.1, 5.2.1.3.2, and 5.2.1.3.3.



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5.2.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2.2 *Packing.*

5.2.2.1 *Level A.*

5.2.2.1.1 *Exterior container.*—The number of units specified in table II shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with the requirements of Federal Specifications PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood, or Military Specifications JAN–P–106, or MIL–B–10377, table II.

5.2.2.1.2 *Case liner.* — Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–B–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.2.1.3 *Strapping.*—Strapping (when required) shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2.2 *Level B.*

5.2.2.2.1 *Exterior container.*—The number of units specified in table II shall be overpacked in an exterior container conforming to the requirements of Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specifications NN–B–621; PPP–B–585, style 3, for class 2 use; or Military Specification MIL–B–10377, table I.

5.2.2.3 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.3 *Marking.*

5.2.3.1 *Level A.*

5.2.3.1.1 *Unit package.*—Each unit package shall be marked with stock number, nomenclature, unit, quantity, lot (control) number, type, color, and name of contractor, and shall also have printed thereon or on a card inclosed therein detailed heat treating instructions, melting range in degree Fahrenheit (F.) and any other instructions for use that the manufacturer deems advisable.

5.2.3.1.2 *Intermediate package.*—Each intermediate package shall be marked with stock number, nomenclature, unit, quantity, lot (control) number, type, color, office identification number, and name of contractor.

5.2.3.1.3 *Protective package and shipping container.* — Each protective package and each shipping container shall be marked in accordance with Military Standard MIL–STD –129, except that no reference to nomenclature nor to the word "gold" shall appear on the shipping container. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments, for the Medical Services of the Armed Forces.

5.2.3.2 *Level B.*

5.2.3.2.1 *Unit package.*—Marking shall be as specified in 5.2.3.1.1.

5.2.3.2.2 *Intermediate package.*—Marking shall be as specified in 5.2.3.1.2.

5.2.3.2.3 *Protective package and shipping container* — Each protective package and each shipping container shall be marked in accordance with Military Standard MIL–STD –129, except that no reference to nomenclature nor to the word "gold" shall appear on the shipping container.



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5.2.4 *Level C.*

5.2.4.1 *Interior packages.*—Each package shall be marked to identify contents.

5.2.4.2 *Shipping containers.* — Shipping containers shall be marked with stock number, name of material, type, color, quantity, name of consignee, name of consignor, and procurement instruction and office identifying numbers as shown on contract or purchase order.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Types.—Requirements given cover only those types of dental gold alloys suitable for casting inlays, other restorations and appliances.

6.3 Uses.—The four types of alloys specified are ordinarily used for the following dental restorations and appliances.

Type I. Those subject to very slight stress and easily burnished.

Type II. Those subject to moderate stress; ¾ crowns, abutments, pontics, full crowns, and sometimes soft saddles.

Type III. Those subject to high stress; thin ¾ crowns, thin cast backings, abutments, pontics, full crowns, and saddles.

Type IV. Those subject to very high stress; saddles, bars, clasps, and unit castings.

6.4 **Hardness number.**—The use of a 13.9 pound (6.3 kg) load for the determination of hardness numbers below 50 is permissible.

6.5 Purchasers should specify type and color required, and should exercise any desired options offered herein. (See 1.2.1, 3.2, 3.4, 5.1.1, 5.1.2, 5.1.3.2, 5.2.1, 5.2.2 and 5.2.3).

6.6 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.7 This specification covers only the types, classes, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and is not intended to include all of the types, etc., which are commercially available.

6.8 **Transportation description.**

6.8.1 Classified as precious metals. Not taken by freight. Ship via Parcel Post or Express.

6.9 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520-580-0250 Gold Alloy, Casting, Dental, Hard, 2 pennyweight.

6520-580-0350 Gold Alloy, Casting, Dental, Soft, 2 pennyweight.

6520-580-0450 Gold Alloy, Casting, Dental, Extra Hard, 2 pennyweight.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto

MILITARY INTERESTS:
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